

Rahul Kumar, PhD

Assistant Professor

Department of Biotechnology, IIT Hyderabad

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EMPLOYMENT

1. Assistant Professor (October 2021 - Present)

Department of Biotechnology, IIT Hyderabad

2. Associate Research Scientist (September 2018 – January 2021)

Institute of Cancer Genetics, Columbia University Irving Medical Center, New York, USA

3. Postdoctoral Fellow (February 2017 – September 2018)

Department of Pathology, Memorial Sloan Kettering Cancer Center (MSKCC), New York, United States.

4. Bioinformatics officer (September 2015 - February 2017)

CRUK Gene Function Laboratory, The Institute of Cancer Research (ICR), London, United Kingdom.

EDUCATIONAL QUALIFICATIONS

1. PhD in Bioinformatics (August 2010 – July 2015)

CSIR-Institute of Microbial Technology (IMTECH), Chandigarh, India.

Thesis Title: “Computational approaches for the designing of anticancer therapy with emphasis on apoptosis pathway”.

Thesis Advisor: Dr. G. P. S. Raghava (www.iiitd.ac.in/raghava)

2. Master of Science (M.Sc.) in Biotechnology (August 2008 – May 2010)

All India Institute of Medical Sciences (AIIMS), New Delhi, India

3. Bachelor of Science (B.Sc.) in Bio-medical Sciences (H) (July 2005 – May 2008)

Acharya Narendra Dev College, University of Delhi, Delhi, India.

RESEARCH WORK AND INTEREST

1. Identification of genetic drivers of cancer progression using computational genomic approaches
2. Application of AI/ML techniques on genomics data to identify diagnostic biomarkers and drug targets identification
3. Integration of radiomics (imaging-based) and multi-omics data to identify non-invasive biomarkers.
4. Deconvolute tumor heterogeneity and mutational processes
5. Identification of drug resistance mechanism in cancer therapies
6. Designing of diagnostic gene panel and personalized cancer therapy
7. Development of biological databases and prediction servers

AWARDS & FELLOWSHIPS

1. Best thesis award for PhD work by CSIR-IMTECH, Chandigarh
2. Faculty Research Excellence Award 2025 by IITH
3. DAAD Visiting Research Fellowship 2024 by DAAD, Germany
4. Scientific High Level Visiting Fellowship 2025 by French Institute in India (IFI), Embassy of France in India

OTHER ACADEMIC ROLES

1. Acting as an **Associate Editor** in nine prestigious science journals i.e., Nature Scientific Reports, BMC Bioinformatics, BMC Cancer, BMC Medical Genomics, Frontiers in Bioinformatics, Frontiers in Big Data, PLoS ONE, PeerJ and Functional & Integrative Genomics.
2. **Core faculty member** in BTech Computational Engineering program since the inception of this program.
3. **Committee member** in Telangana State Council of Higher Education (TSCHE) for designing Bioinformatics curriculum at UG & PG level in Telangana State colleges.

INVITED TALKS

1. Delivered an invited talk on **“Oncogenic expression driven by the hypermutation of super-enhancers in B cell lymphoma”** in 42nd annual conference of Indian Association of Cancer Research 2023 held on Jan 12-16, 2023 in ACTREC Mumbai.
2. Delivered an invited talk on **“AMLdb: A comprehensive multi-omics platform for acute myeloid leukemia”** in 4th annual conference of Indian Cancer Genome Atlas held in IISER Pune on Oct 6-8, 2023.
3. Delivered an invited talk on **“Introduction to Cancer Genetics”** in Rani Durgawati Vishwavidyalaya Jabalpur, MP under UGC-MMTTC program on Aug 29, 2024.
4. Delivered an invited talk on **“Machine Learning Models: From Drug Repurposing. To Predicting Clinical Outcome”** in an international symposium on “Cancer Metabolism and Drug Discovery” organized by Amity University, Noida on Oct 7, 2024.
5. Delivered an invited talk on **“Machine Learning Models: From Drug Repurposing. To Predicting Clinical Outcome”** in “Workshop on Computational Oncology: Modelling and Experiments” organized by Param Hansa Centre for Computations Oncology, IISc on Dec 10, 2024. This event was held at IITH.
6. Delivered an invited talk in a brainstorming session on **“OncoVision: Advancing Together Against Cancer”** between clinicians and scientists held at IIT Bombay on Jan 14, 2025.
7. Online talk in IISER BPR on **“Deciphering the role of super-enhancers in diffuse large B-cell lymphoma”** held on June 11, 2022.
8. Delivered an invited lecture in “Refresher Course on Life Sciences” organized by the **UGC-Human Resource Development Center**, University of Hyderabad on October 20, 2022.
9. Delivered an invited lecture to VIT students on the topic **“Implementation of genomics to understand cancer genetics”** on April 10, 2022.
10. Delivered a lecture and lab visit for school children under **Rural Development Center (RDC)** on April 11, 2025.
11. Presented my teaching and research work and future plans in **48th BoG meeting** of IIT Hyderabad held on Sept 17, 2024.

EVENTS ORGANIZED

1. Organized department's first Industry connect day on September 1, 2025. This event was attended by the 18 industry delegates across pharma and data science companies. Some of them are very interested in having more ties with department and IITH. They are also already offering internship opportunities to our BTech/MTech students.
2. Organized a CCE registered online workshop on title "Computational Methods in Advanced and Precision Oncology" (**7-11th April 2025**), generate a revenue of Rs. 1,20,000.
3. Organizing a CCE registered one-month online workshop on title "BioAI Omics Workshop: Foundations & Practice", scheduled on **November 8-30, 2025**.
4. Hands-on workshop by Dr. Benson Lim, 10X company on Spatial Transcriptomics and Single Cell Sequencing held on **11.03.2025**.
5. Organizing Committee Member in Circadian Rhythms in Health and Diseases: From Discovery to Function (CRHD 2024) held at IIT Hyderabad on **Dec 5-7, 2024**.

RESEARCH PROJECTS

1. **PI:** Investigating the role of MIR122 super-enhancer in hepatocarcinogenesis for therapeutic intervention (DBT), July 2025-July 2028, **40 Lakhs**.
2. **PI:** Unraveling the pathophysiology of metabolic disorders - Proteomics facility (DST-FIST), April 2025-April 2030, **3.5 crore**.
3. **Co-PI:** BRCA1/G-Quadruplex Interactions: Implications in Chromatin Organization and Tumorigenesis (ICMR), April 2025-April 2028, **1.25 crore**.
4. **Co-PI:** Optimized Gene Panel Identification for Kidney Transplant Rejection Using RNA Sequencing and Machine Learning-Based Clustering, Nov 2025- Nov 2028, **1.29 crore**.

PUBLICATIONS

Total Publications	Corresponding Author	#Citations	H-index	i-10 index
69 (22 from IITH)	12 (From IITH)	6385 (As on 31.10.2025)	36	46

Google Scholar Profile: <https://scholar.google.com/citations?user=rDnstZkAAAAJ&hl=en>

#Corresponding author; *Co-first author

Publications from IITH

1. Kavita Kundal, K Divya Rani, Vinodini D, Neeraj Kumar, **Rahul Kumar#** (2025) RadGLO: An interactive platform for radiomic feature analysis and prognostic modeling in glioma. **npj Precision Oncology**, 9(1):323.
2. Simran Jit, Kirti Jain, Leepakshi Dhingra, **Rahul Kumar**, Sherry Bhalla (2025) Alternative Promoters Drive Transcriptomic Reprogramming and Prognostic Stratification in TNBC. **npj Systems Biology and Applications**, 11:121.
3. Kavita Kundal, K Venkateswara Rao, Sandeep Kumar Dhanda, Neeraj Kumar, **Rahul Kumar#** (2025) Tumor Location, Genomic Alterations, and Radiomic Features as Predictors of Survival in Glioblastoma: A Multi-Modal Analysis. **Neuroradiology**; 1-13.
4. Kankana Bhattacharjee, Avik Sengupta, **Rahul Kumar**, Aryya Ghosh (2025). Identification of key hub genes in pancreatic ductal adenocarcinoma: an integrative bioinformatics study. **Frontiers in Bioinformatics**, 5:1536783.
5. Claudio Scuoppo, Bowen Cai, Kenneth Ofori, Hanna Scholze, **Rahul Kumar**, Angelo D'Alessandro, Katia Basso, Laura Pasqualucci, Riccardo Dalla-Favera (2024). Repurposing NAMPT Inhibitors for Germinal Center B Cell–Like Diffuse Large B-Cell Lymphoma. **Blood Cancer Discovery**, 5(6): 417-427.
6. Jyotika Srivastava, Kavita Kundal, Bhuvnesh Rai, Pragati Saxena, Shobhita Katiyar, Naresh Tripathy, Sanjeev Yadav, Ruchi Gupta, **Rahul Kumar**, Soniya Nityanand, & Chandra Prakash Chaturvedi (2024). Global microRNA profiling of bone marrow-MSC derived extracellular vesicles identifies miRNAs associated with hematopoietic dysfunction in aplastic anemia. **Scientific Reports**, 14, 19654.
7. **Rahul Kumar**, Yanfeng Wang, & Sandeep Kumar Dhanda (2024). How well does the adaptive feature representation learning approach identify human mRNA N4-acetylcytidine sites? **Molecular Therapy - Nucleic Acids**, 35, 102300.
8. Kavita Kundal, K Venkateswara Rao, Arunabha Majumdar, Neeraj Kumar, & **Rahul Kumar#** (2024). Comprehensive benchmarking of CNN-based tumor segmentation methods using multimodal MRI data. **Computers in Biology and Medicine**, 178, 108799.
9. Keerthana Vinod Kumar, Ambuj Kumar, Kavita Kundal, Avik Sengupta, Kunjulakshmi R, Subashani Singh, Bhanu Teja Korra, Simran Sharma, Vandana Suresh, Mayilaadumveetil Nishana, & **Rahul Kumar#** (2024). AMLdb: a comprehensive multi-omics platform to identify biomarkers and drug targets for acute myeloid leukemia. **Briefings in Functional Genomics**, elae024.
10. Shuvra Shekhar Roy, Sulochana Bagri, Soujanya Vinayagamurthy, Avik Sengupta, Claudia Regina Then, **Rahul Kumar**, Sriram Sridharan, & Shantanu Chowdhury. (2024). Artificially inserted G-quadruplex DNA secondary structures induce long-distance chromatin activation. **Elife**, 10.7554/eLife.96216.
11. Avik Sengupta, Saurabh Kumar Singh, & **Rahul Kumar#** (2024). Support Vector Machine-Based Prediction Models for Drug Repurposing and Designing Novel Drugs for Colorectal Cancer. **ACS Omega**, 9, 18584-18592.
12. Kunjulakshmi R, Ambuj Kumar, Keerthana Vinod Kumar, Avik Sengupta, Kavita Kundal, Simran Sharma, Ankita Pawar, Pithani Sai Krishna, Mohammad Alfatah, Sandipan Ray, Bhavana Tiwari, & **Rahul Kumar#** (2024). AgingBase: a comprehensive database of anti-aging peptides. **Database**, 2024.

13. Ambuj Kumar, Keerthana Vinod Kumar, Kavita Kundal, Avik Sengupta, Simran Sharma, Kunjulakshmi R, & **Rahul Kumar#** (2024). MyeloDB: a multi-omics resource for multiple myeloma. **Functional & Integrative Genomics**, 24, 17.
14. **Rahul Kumar#**, Kumardeep Chaudhary, & Sandeep Kumar Dhanda (2023). Editorial: Recent advances in peptide informatics: challenges and opportunities. **Frontiers in Bioinformatics**, 3.
15. Sourbh Rankawat, Kavita Kundal, Shreyayukta Chakraborty, **Rahul Kumar**, & Sandipan Ray (2023). A comprehensive rhythmicity analysis of host proteins and immune factors involved in malaria pathogenesis to decipher the importance of host circadian clock in malaria. **Frontiers in Immunology**, 14.
16. Mansi Manchanda, Monica Torres, Farydah Inuossa, Ritu Bansal, **Rahul Kumar**, Matthew Hunt, Craig E. Wheelock, Ety Bachar-Wikstrom, & Jakob D. Wikstrom (2023). Metabolic Reprogramming and Reliance in Human Skin Wound Healing. **Journal of Investigative Dermatology**, 143, 2039-2051.e10.
17. Aswathi Viswanathan, Kavita Kundal, Avik Sengupta, Ambuj Kumar, Keerthana Vinod Kumar, Antony B Holmes, & **Rahul Kumar#** (2023). Deep learning-based classifier of diffuse large B-cell lymphoma cell-of-origin with clinical outcome. **Briefings in Functional Genomics**, 22, 42-48.
18. Vigneshwaran G., Quratulain Annie Hasan, **Rahul Kumar**, & Avinash Eranki (2022). Analysis of single-nucleotide polymorphisms in genes associated with triple-negative breast cancer. **Frontiers in Genetics**, 13.
19. Shwetha Sekar, Yashashwini Subbamanda, Narasimha Pullaguri, Ankush Sharma, Chittaranjan Sahu, **Rahul Kumar**, & Anamika Bhargava (2022). Isoform-specific expression of T-type voltage-gated calcium channels and estrogen receptors in breast cancer reveals specific isoforms that may be potential targets. **Current Research in Biotechnology**, 4, 459-467.
20. Elodie Bal, **Rahul Kumar**, Mohammad Hadigol, Antony B. Holmes, Laura K. Hilton, Jui Wan Loh, Kostiantyn Dreval, Jasper C. H. Wong, Sofija Vlasevska, Clarissa Corinaldesi, Rajesh Kumar Soni, Katia Basso, Ryan D. Morin, Hossein Khiabani, Laura Pasqualucci, & Riccardo Dalla-Favera (2022). Super-enhancer hypermutation alters oncogene expression in B cell lymphoma. **Nature**, 607, 808-815.
21. **Rahul Kumar#**, Youping Deng, Jian-Bing Fan, & Lei Wei (2022). Editorial: Early Detection and Diagnosis of Cancer. **Frontiers in Genetics**, 13.
22. Ambuj Kumar, Keerthana Vinod Kuma, Kunjulakshmi R, Kavita Kundal, Avik Sengupta, & **Rahul Kumar#** (2024). Advancement of in silico tools for stem cell research. In **Computational biology for stem cell research** (pp. 3–16). Elsevier.

Publications before joining IITH

23. Nadeem Riaz, Eric Sherman, Xin Pei, Heiko Schöder, Milan Grkovski, Ramesh Paudyal, Nora Katabi, Pier Selenica, Takafumi N Yamaguchi, Daniel Ma, Simon K Lee, Rachna Shah, **Rahul Kumar**, Fengshen Kuo, Abhirami Ratnakumar, Nathan Aleynick, David Brown, Zhigang Zhang, Vaios Hatzoglou, Lydia Y Liu, Adriana Salcedo, Chiaojung J Tsai, Sean McBride, Luc G T Morris, Jay Boyle, Bhuvanesh Singh, Daniel S Higginson, Rama R Damerla, Arnaud da Cruz Paula, Katharine Price, Eric J Moore, Joaquin J Garcia, Robert Foote, Alan Ho, Richard J Wong, Timothy A Chan, Simon N Powell, Paul C Boutros, John L Humm, Amita Shukla-Dave, David Pfister, Jorge

- S Reis-Filho, & Nancy Lee (2021). Precision Radiotherapy: Reduction in Radiation for Oropharyngeal Cancer in the 30 ROC Trial. **JNCI: Journal of the National Cancer Institute**, 113, 742-751.
24. Mark P. Roberto, Gabriele Varano, Rosa Vinas-Castells, Antony B. Holmes, **Rahul Kumar**, Laura Pasqualucci, Pedro Farinha, David W. Scott, & David Dominguez-Sola (2021). Mutations in the transcription factor FOXO1 mimic positive selection signals to promote germinal center B cell expansion and lymphomagenesis. **Immunity**, 54, 1807-1824.e14.
 25. Yanying Huo, Pier Selenica, Amar H. Mahdi, Fresia Pareja, Kelly Kyker-Snowman, Ying Chen, **Rahul Kumar**, Arnaud Da Cruz Paula, Thais Basili, David N. Brown, Xin Pei, Nadeem Riaz, Yongmei Tan, Yu-Xiu Huang, Tao Li, Nicola J. Barnard, Jorge S. Reis-Filho, Britta Weigelt, & Bing Xia (2021). Genetic interactions among Brca1, Brca2, Palb2, and Trp53 in mammary tumor development. **npj Breast Cancer**, 7, 45.
 26. Marco Fangazio, Erik Ladewig, Karen Gomez, Laura Garcia-Ibanez, **Rahul Kumar**, Julie Teruya-Feldstein, Davide Rossi, Ioan Filip, Qiang Pan-Hammarström, Giorgio Inghirami, Renzo Boldorini, German Ott, Annette M. Staiger, Björn Chapuy, Gianluca Gaidano, Govind Bhagat, Katia Basso, Raul Rabadan, Laura Pasqualucci, & Riccardo Dalla-Favera (2021). Genetic mechanisms of HLA-I loss and immune escape in diffuse large B cell lymphoma. **Proceedings of the National Academy of Sciences**, 118.
 27. Antony B. Holmes, Clarissa Corinaldesi, Qiong Shen, **Rahul Kumar**, Nicolo Compagno, Zhong Wang, Mor Nitzan, Eli Grunstein, Laura Pasqualucci, Riccardo Dalla-Favera, & Katia Basso (2020). Single-cell analysis of germinal-center B cells informs on lymphoma cell of origin and outcome. **Journal of Experimental Medicine**, 217.
 28. **Rahul Kumar**#, & Sandeep Kumar Dhanda (2020). Views on Global Genome Catalogues. **Bioinformation**, 16, 297-300.
 29. Olca Basturk, Britta Weigelt, Volkan Adsay, Jamal K. Benhamida, Gokce Askan, Lu Wang, Maria E. Arcila, Giuseppe Zamboni, Noriyoshi Fukushima, Rodrigo Gularte-Mérida, Arnaud Da Cruz Paula, Pier Selenica, **Rahul Kumar**, Fresia Pareja, Christopher A. Maher, John Scholes, Yoshinao Oda, Donatella Santini, Leona A. Doyle, Iver Petersen, Uta Flucke, Christian Koelsche, Samuel J. Reynolds, Aslihan Yavas, Andreas von Deimling, Jorge S. Reis-Filho, & David S. Klimstra (2020). Sclerosing epithelioid mesenchymal neoplasm of the pancreas – a proposed new entity. **Modern Pathology**, 33, 456-467.
 30. Charles W. Ashley, Arnaud Da Cruz Paula, **Rahul Kumar**, Diana Mandelker, Xin Pei, Nadeem Riaz, Jorge S. Reis-Filho, & Britta Weigelt (2019). Analysis of mutational signatures in primary and metastatic endometrial cancer reveals distinct patterns of DNA repair defects and shifts during tumor progression. **Gynecologic Oncology**, 152, 11-19.
 31. Ana P M Sebastiao, Fresia Pareja, **Rahul Kumar**, David N Brown, Catarina Silveira, Edaise M da Silva, Ju Y Lee, Angela Del, Nora Katabi, Simion Chiosea, Britta Weigelt, Jorge S Reis-Filho, & Raja R Seethala (2019). Genomic analysis of recurrences and high-grade forms of polymorphous adenocarcinoma. **Histopathology**, 75, 193-201.
 32. Anqi Li, Felipe C. Geyer, Pedro Blecua, Ju Youn Lee, Pier Selenica, David N. Brown, Fresia Pareja, Simon S. K. Lee, **Rahul Kumar**, Barbara Rivera, Rui Bi, Salvatore Piscuoglio, Hannah Y. Wen, John R. Lozada, Rodrigo Gularte-Mérida, Luca Cavallone, Zoulikha Rezoug, Tu Nguyen-Dumont, Paolo Peterlongo, Carlo Tondini, Thorkild Terkelsen, Karina Rønlund, Susanne E. Boonen, Arto Mannerma, Robert Winqvist, Marketa Janatova, Pathmanathan Rajadurai, Bing Xia, Larry Norton, Mark E. Robson, Pei-Sze Ng, Lai-Meng Looi, Melissa C. Southey, Britta Weigelt, Teo

- Soo-Hwang, Marc Tischkowitz, William D. Foulkes, Jorge S. Reis-Filho (2019). Homologous recombination DNA repair defects in PALB2-associated breast cancers. **npj Breast Cancer**, 5, 23.
33. Snjezana Dogan, Charlotte K.Y. Ng, Bin Xu, **Rahul Kumar**, Lu Wang, Marcia Edelweiss, Sasinya N. Scott, Ahmet Zehir, Alexander Drilon, Luc G.T. Morris, Nancy Y. Lee, Cristina R. Antonescu, Alan L. Ho, Nora Katabi, Michael F. Berger, & Jorge S. Reis-Filho (2019). The repertoire of genetic alterations in salivary duct carcinoma including a novel HNRNPH3-ALK rearrangement. **Human Pathology**, 88, 66-77.
 34. Fresia Pareja, Ju Youn Lee, David N Brown, Salvatore Piscuoglio, Rodrigo Gularte-Mérida, Pier Selenica, Arnaud Da Cruz Paula, Sasi Arunachalam, **Rahul Kumar**, Felipe C Geyer, Catarina Silveira, Edaise M da Silva, Anqi Li, Caterina Marchiò, Charlotte K Y Ng, Odette Mariani, Laetitia Fuhrmann, Hannah Y Wen, Larry Norton, Anne Vincent-Salomon, Edi Brogi, Jorge S Reis-Filho, & Britta Weigelt (2019). The Genomic Landscape of Mucinous Breast Cancer. **JNCI: Journal of the National Cancer Institute**, 111, 737-741.
 35. Pier Selenica*, Nitya Raj*, **Rahul Kumar***, David N. Brown, Oriol Arqués, Diane Reidy, David Klimstra, Matija Snuderl, Jonathan Serrano, Héctor G. Palmer, Britta Weigelt, Jorge S. Reis-Filho, & Maurizio Scaltriti (2019). Solid pseudopapillary neoplasms of the pancreas are dependent on the Wnt pathway. **Molecular Oncology**, 13, 1684-1692.
 36. Diana Mandelker*, **Rahul Kumar***, Xin Pei, Pier Selenica, Jeremy Setton, Sasi Arunachalam, Ozge Ceyhan-Birsoy, David N Brown, Larry Norton, Mark E Robson, Hannah Y Wen, Simon Powell, Nadeem Riaz, Britta Weigelt, & Jorge S Reis-Filho (2019). The Landscape of Somatic Genetic Alterations in Breast Cancers from CHEK2 Germline Mutation Carriers. **JNCI Cancer Spectrum**, 3.
 37. Felipe C Geyer, Anqi Li, Anastasios D Papanastasiou, Alison Smith, Pier Selenica, Kathleen A Burke, Marcia Edelweiss, Huei-Chi Wen, Salvatore Piscuoglio, Anne M Schultheis, Luciano G Martelotto, Fresia Pareja, **Rahul Kumar**, Alissa Brandes, Dan Fan, Thais Basili, Arnaud Da Cruz Paula, John R Lozada, Pedro Blequa, Simone Muenst, Achim A Jungbluth, Maria P Foschini, Hannah Y Wen, Edi Brogi, Juan Palazzo, Brian P Rubin, Charlotte K Y Ng, Larry Norton, Zsuzsanna Varga, Ian O Ellis, Emad A Rakha, Sarat Chandralapaty, Britta Weigelt, & Jorge S Reis-Filho (2018). Recurrent hotspot mutations in HRAS Q61 and PI3K-AKT pathway genes as drivers of breast adenomyoepitheliomas. **Nature Communications**, 9, 1816.
 38. Jennifer J Mueller, Brooke A Schlappé, **Rahul Kumar**, Narciso Olvera, Fanny Dao, Nadeem Aburustum, Carol Aghajanian, Deborah DeLair, Yaser R Hussein, Robert A Soslow, Douglas A Levine, & Britta Weigelt (2018). Massively parallel sequencing analysis of mucinous ovarian carcinomas: genomic profiling and differential diagnoses. **Gynecologic Oncology**, 150, 127-135.
 39. Rachel Brough, Aditi Gulati, Syed Haider, **Rahul Kumar**, James Campbell, Erik Knudsen, Stephen J Pettitt, Colm J Ryan, & Christopher J Lord (2018). Identification of highly penetrant Rb-related synthetic lethal interactions in triple negative breast cancer. **Oncogene**, 1.
 40. Fresia Pareja, Alissa H Brandes, Thais Basili, Pier Selenica, Felipe C Geyer, Dan Fan, Arnaud Da Cruz Paula, **Rahul Kumar**, David N Brown, Rodrigo Gularte-Mérida, Barbara Alemar, Rui Bi, Raymond S Lim, Ino de Bruijn, Sho Fujisawa, Rui Gardner, Elvin Feng, Anqi Li, Edaise M da Silva, John R Lozada, Pedro Blequa, Leona Cohen-Gould, Achim A Jungbluth, Emad A Rakha, Ian O Ellis, Maria I A Edelweiss, Juan Palazzo, Larry Norton, Travis Hollmann, Marcia Edelweiss, Brian P Rubin, Britta Weigelt, & Jorge S Reis-Filho (2018). Loss-of-function mutations in ATP6AP1 and ATP6AP2 in granular cell tumors. **Nature Communications**, 9, 3533.
 41. Ömer Sinan Saraç, **Rahul Kumar**, Sandeep Kumar Dhanda, Ali Tuğrul Balcı, İsmail Bilgen, Roberto Romero, & Adi L Tarca (2018). Species translatable blood gene signature as a marker of

- exposure to smoking: Computational approaches of the top ranked teams in the sbv IMPROVER Systems Toxicology Challenge. **Computational Toxicology**, 5, 25-30.
42. Vincenzo Belcastro, Carine Poussin, Yang Xiang, Maurizio Giordano, Kumar Parijat Tripathi, Akash Boda, Ali Tugrul Balci, Ismail Bilgen, Sandeep Kumar Dhanda, Zhongqu Duan, Xiaofeng Gong, **Rahul Kumar**, Roberto Romero, Omer Sinan Sarac, Adi L Tarca, Peixuan Wang, Hao Yang, Wenxin Yang, Chenfang Zhang, Stéphanie Boué, Mario Rosario Guarracino, Florian Martin, Manuel C Peitsch, & Julia Hoeng (2018). The sbv IMPROVER Systems Toxicology computational challenge: Identification of human and species-independent blood response markers as predictors of smoking exposure and cessation status. **Computational Toxicology**, 5, 38-51.
 43. Jisun Kim, Felipe C Geyer, Luciano G Martelotto, Charlotte K Y Ng, Raymond S Lim, Pier Selenica, Anqi Li, Fresia Pareja, Nicola Fusco, Marcia Edelweiss, **Rahul Kumar**, Rodrigo Gularte-Merida, Andre N Forbes, Ekta Khurana, Odette Mariani, Sunil Badve, Anne Vincent-Salomon, Larry Norton, Jorge S Reis-Filho, & Britta Weigelt (2018). MYBL1 rearrangements and MYB amplification in breast adenoid cystic carcinomas lacking the MYB - NFIB fusion gene. **The Journal of Pathology**, 244, 143-150.
 44. Britta Weigelt*, Rui Bi*, **Rahul Kumar***, Pedro Blecua, Diana L Mandelker, Felipe C Geyer, Fresia Pareja, Paul A James, Fergus J Couch, Diana M Eccles, Fiona Blows, Paul Pharoah, Anqi Li, Pier Selenica, Raymond S Lim, Gowtham Jayakumaran, Nic Waddell, Ronglai Shen, Larry Norton, Hannah Y Wen, Simon N Powell, Nadeem Riaz, Mark E Robson, Jorge S Reis-Filho, & Georgia Chenevix-Trench (2018). The Landscape of Somatic Genetic Alterations in Breast Cancers From ATM Germline Mutation Carriers. **JNCI: Journal of the National Cancer Institute**.
 45. Ilirjana Bajrami, Rebecca Marlow, Marieke van de Ven, Rachel Brough, Helen N Pemberton, Jessica Frankum, Feifei Song, Rumana Rafiq, Asha Konde, Dragomir B Krastev, Malini Menon, James Campbell, Aditi Gulati, **Rahul Kumar**, Stephen J Pettitt, Mark D Gurden, Marta Llorca Cardenosa, Irene Chong, Patrycja Gazinska, Fredrik Wallberg, Elinor J Sawyer, Lesley-Ann Martin, Mitch Dowsett, Spiros Linardopoulos, Rachael Natrajan, Colm J Ryan, Patrick W B Derksen, Jos Jonkers, Andrew N J Tutt, Alan Ashworth, & Christopher J Lord (2018). E-Cadherin/ROS1 Inhibitor Synthetic Lethality in Breast Cancer. **Cancer Discovery**, 8, 498-515.
 46. Jenni Nikkilä, **Rahul Kumar**, James Campbell, Inger Brandsma, Helen N Pemberton, Fredrik Wallberg, Kinga Nagy, Ildikó Scheer, Beata G Vertessy, Artur A Serebrenik, Valentina Monni, Reuben S Harris, Stephen J Pettitt, Alan Ashworth, & Christopher J Lord (2017). Elevated APOBEC3B expression drives a kataegic-like mutation signature and replication stress-related therapeutic vulnerabilities in p53-defective cells. **British Journal of Cancer**, 117, 113-123.
 47. Fresia Pareja, Felipe C Geyer, **Rahul Kumar**, Pier Selenica, Salvatore Piscuoglio, Charlotte K Y Ng, Kathleen A Burke, Marcia Edelweiss, Melissa P Murray, Edi Brogi, Britta Weigelt, & Jorge S Reis-Filho (2017). Phyllodes tumors with and without fibroadenoma-like areas display distinct genomic features and may evolve through distinct pathways. **NPJ breast cancer**, 3, 40.
 48. Sudheer Gupta, Kumardeep Chaudhary, Sandeep Kumar Dhanda, **Rahul Kumar**, Shailesh Kumar, Manika Sehgal, Gandharva Nagpal, & Gajendra P. S. Raghava (2016). A Platform for Designing Genome-Based Personalized Immunotherapy or Vaccine against Cancer. **PLOS ONE**, 11, e0166372.
 49. Jagpreet Singh Nanda*, **Rahul Kumar***, & Gajendra P S Raghava (2016). dbEM: A database of epigenetic modifiers curated from cancerous and normal genomes. **Scientific reports**, 6, 19340.
 50. Sudheer Gupta, Kumardeep Chaudhary, **Rahul Kumar**, Ankur Gautam, Jagpreet Singh Nanda, Sandeep Kumar Dhanda, Samir Kumar Brahmachari, & Gajendra P S Raghava (2016).

- Prioritization of anticancer drugs against a cancer using genomic features of cancer cells: A step towards personalized medicine. **Scientific reports**, 6, 23857.
51. **Rahul Kumar**, & Gajendra P S Raghava (2016). ApoCanD: Database of human apoptotic proteins in the context of cancer. **Scientific reports**, 6, 20797.
 52. **Rahul Kumar**, Jagat Chauhan, & Gajendra Raghava (2015). In Silico Designing and Screening of Antagonists against Cancer Drug Target XIAP. **Current Cancer Drug Targets**, 15, 836-846.
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WEB SERVICES DEVELOPED AT IIT Hyderabad

We have developed following prediction web servers and databases at IITH.

1. COLORECPRED: <https://project.iith.ac.in/cgntlab/colorecpred/>
2. AagingBase: <https://project.iith.ac.in/cgntlab/aagingbase/>
3. MyeloDB: <https://project.iith.ac.in/cgntlab/myelodb/>
4. AMLdb: <https://project.iith.ac.in/cgntlab/amldb/>
5. RadGLO: <https://project.iith.ac.in/cgntlab/radglo/>
6. MLP-COO: <https://github.com/CGnTLab/MLP-COO>
7. tumorVQ: <https://github.com/CGnTLab/tumorVQ>
8. eVGeMdb: <https://project.iith.ac.in/cgntlab/eVGeMdb/>

9. AGEX: <https://project.iith.ac.in/cgntlab/AGEX/>

10. Plasmorhythm: <https://project.iith.ac.in/cgntlab/Plasmorhythm/>

STUDENTS

PhD Ongoing	PhD Graduated	MTech Ongoing	MTech Graduated
7	Open colloquium (OC) of Ms. Kavita Kundal (BO22RESCH01003) was conducted on October 29, 2025. PhD viva-voce will be conducted in December 2025.	2	5